INTERNSHIP REPORT

Submitted

For the Partial Fulfilment of the Requirement for the Degree of Bachelor of _____ with _____ (Major and Minor Subjects) under **Magadh University**

Submitted to

The Department of _____, _____ **(College Name)**

[Paste College Logo here]

Submitted by

Name : _____

Semester : _____

Programme FYUGP : _____

University's Registration No. : _____

College Roll No. : _____

Year : _____

RK NEXORA
CAMPUS TO CAREER
A Unit of RKNexora Private Limited.

INTERNSHIP REPORT

Topic - Artificial Intelligence

Submitted to the Head of Department, _____ (College Name), in partial fulfilment of the requirements for the Degree of _____, Magadh University

Name of Student : _____

Programme : _____

University Registration No. : _____

College Roll No. : _____

Internship Registration No. : _____

Name of the IPO / Agency : RKNEXORA PRIVATE LIMITED

Internship Period : _____ to _____

Semester : _____

Year / Session : _____

RKNEXORA PRIVATE LIMITED
RKNEXORA
PRIVATE LIMITED
GAYA, BIHAR - 805236, INDIA

_____
Signature of Student

_____
Seal & Signature of IPO

ACKNOWLEDGEMENT

I express my sincere gratitude to RK NEXORA, RKNEXORA PRIVATE LIMITED, _____ (College Name), my faculty mentors and the internship supervisor, _____, for their guidance and support. The structured activities, learning resources and regular feedback helped me complete this internship in Artificial Intelligence successfully.

I am thankful to the Head of the Department and the college mentor for their encouragement and for guiding me in preparing this report. I am also grateful to the management of RKNEXORA PRIVATE LIMITED for providing a professional learning environment and the opportunity to learn and apply my knowledge in a practical setting. The concept sessions, hands-on coding tasks with sample datasets and free tools, and the supervisor's patient and detailed feedback helped me understand how intelligent systems are built, tested and used responsibly behind the scenes.

Finally, I thank my family and peers for their constant encouragement throughout the programme.

_____

Name: _____

University Registration No. - _____

INTERNSHIP COMPLETION CERTIFICATE

The Internship Completion Certificate issued by RKNEXORA PRIVATE LIMITED for the internship in Artificial Intelligence is to be attached on this page.

[Attach the original / printed copy of the QR-verified Internship Completion Certificate here]

Certificate No. _____

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PREFACE

Internships are an important part of the undergraduate programme, as they allow students to see how classroom learning is applied in real organisations. The Four-Year Undergraduate Programme (FYUGP), introduced in line with the National Education Policy 2020, gives special importance to experiential learning, skill development and employability. This report has been prepared as part of the internship requirement of Magadh University and the Higher Education guidelines applicable in Bihar.

The report describes the internship undertaken in Artificial Intelligence at RKNEXORA PRIVATE LIMITED, the plan followed, the work carried out, and the learning gained. Today most people meet Artificial Intelligence (AI) many times a day without noticing it: the phone suggests the next word, a map predicts the travel time, a bank flags a suspicious transaction, a video app recommends what to watch, and a voice assistant understands a spoken question in Hindi or English. Businesses of every size, from a small shop in Gaya to a national company, are therefore asking how data can be turned into better decisions. AI is the set of methods that make this possible: programming, data preparation, machine learning, neural networks, language processing and computer vision. This internship offered a first-hand introduction to how these methods work together.

The report has been arranged in the order suggested by the University, beginning with the cover page and acknowledgement and ending with the conclusion and attendance sheet. It is hoped that the report gives a clear and honest account of the internship experience. All examples in it use dummy data and fictitious organisations; no real personal data was handled and no confidential dataset was used.

EXECUTIVE SUMMARY

This report documents the internship completed in Artificial Intelligence at RKNEXORA PRIVATE LIMITED (RK NEXORA - 'Campus to Career'). The programme required a minimum of 120 hours, completed over about four to five weeks (28 working days) with daily sessions of four to five hours, and was supported by an attendance log, supervisor guidance and review.

Background. Artificial Intelligence means building computer systems that perform tasks that normally need human intelligence, such as learning from examples, recognising patterns, understanding language and images, and making predictions. Its great strength is that a system can improve from data, and its quality can be measured: how often a model is right, what kinds of mistakes it makes, and whether it works on new data. A good AI practitioner therefore combines programming, mathematics and careful thinking with an awareness of ethics. For a student from any discipline, the internship shows how logic, statistics, language, technology and problem-solving come together in a real AI job.

Objectives. The internship aimed to (i) build a structured understanding of AI fundamentals and the difference between AI, machine learning and deep learning, (ii) learn Python, NumPy and Pandas for AI work, (iii) learn data preprocessing, (iv) learn supervised and unsupervised learning, (v) learn model evaluation, neural network basics, NLP basics and computer vision basics, and (vi) develop communication, documentation and professional discipline.

Work carried out. The student first studied AI fundamentals, then worked through Python for AI, data preprocessing, supervised learning, unsupervised learning, model evaluation, neural network basics, NLP basics and computer vision basics. Each topic had a concept session followed by a hands-on task using free tools, open sample datasets and dummy data. The later part of the programme was devoted to a mini AI project, a case study, a mock practical exercise, supervisor review and preparation of this report.

Project. The work was tied together in a project titled *AI Mini Project using Machine Learning*, in which data was prepared, a small model was built and tested, and its results were evaluated and reported.

Method. A blended method was followed - learn, practise, document and review - supported by a daily attendance log with day-wise remarks and regular feedback from the supervisor.

Learning gained. The student gained a clear understanding of how a small machine learning solution is planned, built, tested and explained; practical familiarity with Python, NumPy, Pandas, data cleaning, encoding and scaling, regression and classification, clustering, evaluation metrics, a simple neural network, text processing and image processing; awareness of bias, privacy, consent and data protection rules; and improved communication, analytical thinking and professional discipline.

Key observations. Good AI starts with a clear problem and good data; a model is only as good as the data it learns from; accuracy alone can be misleading; a model must be tested on data it has never seen; and results should lead to decisions and not merely list numbers.

Conclusion. The internship met its objectives and contributed to the student's academic understanding, employability and confidence. The report follows the University's suggestive format: cover page, acknowledgement, certificate, contents, preface, executive summary, about the agency, plan and execution, learning outcome, conclusion and attendance sheet.

ABOUT THE AGENCY

RK NEXORA, a unit of RKNEXORA PRIVATE LIMITED, provides structured internship and skill-development programmes under the theme 'Campus to Career'. The organization focuses on academic-industry alignment, practical exposure, assessment, documentation and employability-oriented learning.

Particulars	Details
Name of Agency	RKNEXORA PRIVATE LIMITED
Programme / Unit	RK NEXORA - Campus to Career
Registered Office	C/O Mukesh Kumar, Khiri, PS Atri, Jethiyan, Atri, Gaya - 805236, Bihar, India
CIN	U85499BR2026PTC086757
Email	rknexorapvtltd@gmail.com
Website	www.rknexora.org
Internship Topic	Artificial Intelligence

1. Vision

To support students in developing practical, professional and industry-relevant capabilities.

2. Mission

To deliver structured learning experiences that connect academic knowledge with workplace expectations.

3. Activities of the Agency

1. Structured internship programmes aligned with university and higher-education requirements.
2. Skill-development learning modules, practical activities and assignments.
3. Attendance tracking, supervision, assessment and documentation.
4. Employability-oriented guidance connecting campus learning to career readiness.

4. Programme Structure

The internship programme is organised around a fixed number of learning hours, completed over a defined period. Each intern is registered on the internship portal with a unique registration number, is assigned a supervisor, follows a day-wise plan and maintains an attendance log. Learning is organised into three phases: **Orientation** (Days 1-2), **Learning and Practice** (Days 3-22, ten sub-topics of two days each: one concept day and one hands-on day) and **Project and Review** (Days 23-28), followed by a concluding review.

5. Role of the Supervisor

The internship supervisor assigned by the agency guided the student through the programme, explained concepts, assigned practice tasks, reviewed the daily log and gave feedback. The supervisor also verified attendance and reviewed the student's work at the end of the programme.

PLAN AND EXECUTION

1. Objectives

The internship was planned around the following objectives, each linked to a part of the programme:

1. Understand AI fundamentals - what AI is, its types, applications, and the difference between AI, ML and DL (Week 1).
2. Learn Python for AI - Python basics, NumPy and Pandas used in AI work (Week 1).
3. Learn data preprocessing - cleaning data, handling missing values, encoding and scaling (Week 2).
4. Learn supervised and unsupervised learning - regression and classification with labelled data; clustering, dimensionality reduction and pattern finding (Week 2).
5. Learn model evaluation - accuracy, precision, recall, confusion matrix and overfitting (Week 3).
6. Learn neural network basics, NLP basics and computer vision basics - neurons, layers, activation functions and training; text processing and tokenisation; image data and basic image processing (Weeks 3-4).
7. Complete a mini AI project - planning, building and testing a small AI model end to end (Week 4).
8. Develop professional discipline - communication, honesty, documentation and report writing (throughout).

2. Methodology

The programme followed a blended method made up of four connected steps, repeated for each topic:

Learn. Each topic began with a concept session in which the supervisor explained the purpose of the technique or tool, its main terms and how it connects with the other parts of an AI workflow. Notes were prepared in the student's own words.

Practise. The concepts were then applied in guided hands-on tasks - writing small Python programs, exploring a dataset with Pandas, cleaning a messy table, training a regression and a classification model, clustering sample customers, calculating metrics from a confusion matrix, tracing a single neuron by hand, tokenising sentences and treating a small image as an array of numbers. Only open sample datasets, dummy data and free tools such as Jupyter Notebook or Google Colab were used, so that no real personal data was handled.

Document. Every day's work was recorded in the internship attendance log with a short remark, and notebooks, datasets and notes were saved so that the work could be reviewed.

Review. The supervisor reviewed the work, pointed out errors and suggested improvements. Review days, a case study and a mock practical exercise helped to consolidate the learning.

3. Understanding Artificial Intelligence - Concepts Studied

3.1 AI fundamentals

Artificial Intelligence is the field of computer science concerned with building systems that perform tasks that normally require human intelligence, such as learning, reasoning, perception, language understanding and decision making. The idea of machine intelligence was discussed by Alan Turing in 1950, and the term 'Artificial Intelligence' was introduced at the Dartmouth workshop in 1956. Since then the field has moved from hand-written rules to systems that learn from data.

AI is often described by capability. *Narrow AI* does one task well, such as spam filtering or translation, and all AI in use today is of this kind. *General AI*, a system with human-level ability across tasks, remains a research goal and does not exist today. It is also common to describe systems by how they work: reactive

systems respond only to the present input, while systems with memory use past data to improve their decisions.

Three terms are often confused and are best understood as nested circles. **Artificial Intelligence** is the widest idea - any technique that lets a machine behave intelligently, including rule-based systems and search. **Machine Learning (ML)** is a part of AI in which a system learns patterns from data instead of being given explicit rules. **Deep Learning (DL)** is a part of ML that uses neural networks with many layers and is behind most modern image, speech and language systems. *Generative AI* refers to models that create new text, images or code; it is powerful but can produce confident mistakes (often called hallucinations), so its output must be checked.

The main kinds of learning are supervised learning (learning from labelled examples), unsupervised learning (finding structure in unlabelled data) and reinforcement learning (learning by reward and penalty). Applications are wide: disease and crop-disease detection from images, fraud detection in banking, recommendation in retail and media, adaptive learning in education, speech and translation tools for Indian languages, and demand forecasting in supply chains. A useful habit taught in the internship is to begin every AI task by asking: *what is the problem, what data is available, how will success be measured, and what could go wrong?*

3.2 Python for AI

Python is the most widely used language for AI because it is easy to read and has a very large collection of libraries. The basics studied were variables and data types (integers, floats, strings, booleans), lists, tuples, dictionaries and sets, conditions and loops, functions, reading and writing files, handling errors and importing libraries. Programs were written in Jupyter Notebook or Google Colab, which mix code, results and notes in one place and make work easy to review.

NumPy provides the ndarray, a fast multi-dimensional array. Its strength is *vectorised* operations: an operation applied to a whole array at once, without writing a loop. Operations practised were creating arrays, checking shape and data type, reshaping, indexing and slicing, broadcasting (adding a number or a row to every element), and statistics such as mean, median, standard deviation, sum, minimum and maximum, along rows or columns.

Pandas provides the Series and the DataFrame, which behave like columns and tables. Operations practised were reading a CSV file, viewing the first rows (head), checking columns and types (info), summarising numbers (describe), selecting columns and rows (loc and iloc), filtering by conditions, sorting, counting missing values, grouping and summarising (groupby), merging tables and saving the result. **Matplotlib** was used for line charts, bar charts, histograms and scatter plots, because a chart often shows a problem in the data faster than a table does.

Good coding practice was stressed: meaningful variable names, short comments, small functions, running cells in order, reading the error message from the last line upwards, keeping the raw data file unchanged and saving cleaned data separately, and fixing random seeds so that results can be repeated.

3.3 Data preprocessing

Real data is rarely ready for a model. Data preprocessing is the work of turning raw data into a clean, consistent table that a model can learn from, and in practice it takes more time than building the model. The work begins with *exploration*: checking the number of rows and columns, the type of each column, the range of values, and charts of each feature, so that problems are seen before they cause wrong results.

Missing values may be handled by removing rows or columns (only when little is lost), or by filling with the mean or median for numbers and the most frequent value (mode) for categories; the median is safer when there are extreme values. **Duplicates** and wrong data types (for example numbers stored as text) are removed or corrected. **Outliers** are values far from the rest and are found with charts, the interquartile range rule or

z-scores; they are not deleted automatically, because an outlier may be a real and important case.

Encoding converts categories into numbers. Label encoding gives each category a number and suits ordered categories; one-hot encoding creates one 0/1 column per category and suits unordered ones such as district or subject. **Scaling** puts numbers on a comparable range: min-max scaling gives (x − min) / (max − min), which lies between 0 and 1, and standardisation gives (x − mean) / standard deviation. Scaling matters for distance-based and neural network models. Finally the data is split into a *training set* (the model learns from it), an optional *validation set* (for choosing settings) and a *test set* (kept aside for the final check), commonly in the ratio 80:20 or 70:15:15.

Two cautions were stressed. First, **data leakage**: any step that learns from data, such as the mean used for filling or the scaler, must be learned from the training set only and then applied to the test set; otherwise the test result is falsely good. Second, **class imbalance**: if one class is very rare, the data may need stratified splitting, resampling or different metrics. Feature engineering - creating a useful new column, such as 'attendance percentage' from days present and days held - often helps more than a complicated model.

3.4 Supervised learning

In supervised learning every training example has an input (features) and a known answer (label), and the model learns the relationship between them. It has two main forms. **Regression** predicts a number, such as marks, price or demand. **Classification** predicts a category, such as pass or fail, spam or not spam, or disease or no disease.

Simple linear regression fits a straight line $y = m\,x + c$ through the points by minimising the squared errors; with several features it becomes multiple linear regression. Regression is judged by errors such as MAE, MSE, RMSE and by R^2, which shows how much of the variation the model explains. For classification the models studied were logistic regression (which gives a probability), k-nearest neighbours (which votes among the closest examples), decision trees (which ask a series of yes/no questions) and random forests (many trees combined). Naive Bayes and support vector machines were introduced briefly.

The standard workflow is: define the target, prepare the data, split into train and test, fit the model on the training data, predict on the test data and evaluate. The central difficulty is **generalisation**. A model that is too simple *underfits* and is poor even on training data; a model that is too complex *overfits*, memorising the training data and failing on new data. Remedies include more data, simpler models, limiting tree depth, regularisation, cross-validation (testing on several different splits) and tuning *hyper-parameters*, which are settings chosen by the human, such as the value of k or the depth of a tree.

3.5 Unsupervised learning

In unsupervised learning there are no labels, and the aim is to discover structure. **Clustering** groups similar items. In *k-means* the number of clusters k is chosen; k starting centres are placed; each point joins its nearest centre; each centre moves to the mean of its points; and the last two steps repeat until nothing changes. The value of k can be guided by the *elbow method* (plotting the total within-cluster distance for different k) and the *silhouette score*. Hierarchical clustering and density-based clustering (DBSCAN, which also finds outliers) were introduced. Scaling is essential before clustering, because a feature with large numbers would otherwise dominate the distance.

Dimensionality reduction reduces many features to a few while keeping most of the information; *Principal Component Analysis (PCA)* creates new axes along the directions of greatest variation and is used to visualise high-dimensional data in two dimensions and to remove noise. **Pattern finding** includes association rules, as in market-basket analysis ('customers who buy A often buy B'), and anomaly detection, which flags unusual transactions or sensor readings. Typical uses are customer segmentation, grouping similar documents, and spotting fraud. Because there is no correct answer to compare with, clusters must be checked for usefulness and explained in words, and different settings may give different groupings.

3.6 Model evaluation

A model is useful only if it works on new data, so evaluation is as important as building. For classification the **confusion matrix** counts true positives (TP), false positives (FP), false negatives (FN) and true negatives (TN). From it: accuracy = (TP + TN) / total; precision = TP / (TP + FP), which asks 'when the model says yes, how often is it right?'; recall = TP / (TP + FN), which asks 'of all real yes cases, how many were found?'; and F1 = 2 × precision × recall / (precision + recall), a balance of the two. The ROC curve and AUC show performance across all thresholds.

Accuracy alone can mislead. If only 5% of loan applicants default, a model that predicts 'no default' for everyone is 95% accurate and completely useless. Which metric matters depends on the cost of each error: in disease screening a missed case (low recall) is more serious, while in a spam filter a good mail wrongly blocked (low precision) is more serious. For regression the metrics are MAE, MSE, RMSE and R^2.

Overfitting is recognised when training performance is high and test performance is much lower; **underfitting** when both are low. Good practice includes comparing with a simple *baseline*, using cross-validation, keeping the test set untouched until the end, reading examples of wrong predictions (error analysis), and checking whether the model performs equally well for different groups of people.

3.7 Neural network basics

A neural network is a model made of simple units called neurons, loosely inspired by the brain. A neuron multiplies each input by a weight, adds them with a bias, and passes the sum z through an *activation function*: $z = w_1x_1 + w_2x_2 + \ldots + b$, output = f(z). Common activations are sigmoid (output between 0 and 1), tanh, ReLU (output is z if positive, otherwise 0) and softmax (turns scores into probabilities over classes). Without a non-linear activation, many layers would collapse into one linear model.

Neurons are arranged in an *input layer*, one or more *hidden layers* and an *output layer*. Training repeats four steps: a forward pass produces a prediction; a *loss function* measures the error; *backpropagation* works out how much each weight contributed to the error; and *gradient descent* adjusts the weights a little in the direction that reduces the loss. Important settings are the learning rate (size of each step), the number of epochs (passes over the data) and the batch size. Neural networks overfit easily, so dropout, early stopping and more data are used. The internship introduced Keras/TensorFlow at a basic level and gave an overview of convolutional networks (for images), recurrent networks (for sequences) and transformers (behind modern language models). Neural networks need more data and computing power and are harder to explain, so a simpler model is preferred when it performs well enough.

3.8 NLP basics

Natural Language Processing (NLP) enables computers to work with human language. A typical text pipeline is: collect the text, convert to lower case, remove punctuation and extra spaces, **tokenise** (split into words or sub-words), remove common *stop words* such as 'is' and 'the' where useful, and apply *stemming* (cutting words to a root) or *lemmatisation* (reducing to a dictionary form). Text is then turned into numbers: *bag-of-words* counts each word, *TF-IDF* gives more weight to words that are frequent in one document but rare overall, and *embeddings* represent words as vectors so that similar meanings lie close together.

Simple NLP tasks practised were sentiment classification of short reviews, word-frequency counting, keyword extraction and a basic spam filter; larger tasks introduced were named entity recognition, machine translation, summarisation, question answering and chatbots. Modern large language models are built on the transformer architecture and are trained on very large text collections. Language in India adds challenges: many languages and scripts, Hinglish typed in Roman letters, spelling variations and limited data for some languages. Models can reflect bias in their training text and can state false facts fluently, so their output needs human checking.

3.9 Computer vision basics

Computer vision enables machines to understand images and video. To a computer an image is an array of numbers: a grayscale image is a 2-D grid of pixel values from 0 (black) to 255 (white), and a colour image has three channels (red, green, blue). Basic processing studied was loading and displaying an image, resizing, converting to grayscale, normalising pixel values to 0-1, blurring to reduce noise, thresholding, edge detection and cropping, using Pillow, OpenCV, NumPy and Matplotlib.

A *filter* (kernel) is a small grid that slides over the image; for example a 3×3 averaging filter replaces each pixel by the mean of its neighbours and blurs the image. Convolutional neural networks learn their own filters, with early layers detecting edges and later layers detecting shapes and objects, and *pooling* layers reduce the size. Main tasks are image classification, object detection, segmentation and text recognition (OCR). Data augmentation (flips, small rotations, brightness changes) creates more training variety. Uses include crop-disease detection, medical image support, traffic and quality inspection. Face recognition and surveillance raise serious privacy questions, so consent, purpose and law must be considered.

3.10 Mini AI project - building a model end to end

The mini project joins all earlier topics into one workflow: (1) define the problem and success measure; (2) collect or choose a dataset and note its source and licence; (3) explore the data; (4) clean and prepare it; (5) split into train and test; (6) build a simple baseline and then one or two models; (7) evaluate on the test set; (8) improve and compare; (9) explain the results and limits in plain language; and (10) save the notebook, data description and a short 'model card' stating what the model does, what data it used, how well it works and where it should not be used. Version control and clear notes make the work repeatable, and the project report must state assumptions and failures honestly.

3.11 Rules, ethics and limitations

AI work must follow laws and ethical principles. The Digital Personal Data Protection Act, 2023 requires lawful, purpose-limited and consent-based use of personal data. The Information Technology Act, 2000 governs electronic records and online conduct, and the Copyright Act, 1957 applies to datasets, code, text and images used for training and to generated content. India's policy papers on responsible AI and international guidance such as the UNESCO Recommendation on the Ethics of AI (2021) stress fairness, transparency, accountability, safety, privacy and human oversight.

Good practice includes using data only with permission and for a stated purpose, removing or masking names and identifiers, checking datasets for bias and for missing groups, testing performance across groups, explaining model decisions where people are affected, keeping a human in the loop for important decisions, crediting data and code sources, and not claiming more accuracy than was measured. The tools, library versions and rules mentioned in this report change often, so they must be verified against official documentation, and the figures in the examples are dummy values for learning only.

4. Week-wise Plan and Execution

Week	Days	Activities
Week 1	Days 1-6	Orientation and foundations: company induction, introduction to Artificial Intelligence, AI fundamentals, and Python for AI (Python basics, NumPy and Pandas).
Week 2	Days 7-12	Prepare and learn: data preprocessing, supervised learning and unsupervised learning, each with a concept session and a hands-on task.
Week 3	Days 13-18	Evaluate and build: model evaluation, neural network basics and NLP basics.
Week 4	Days 19-24	Vision and project: computer vision basics, mini AI project, a case study exercise and a mock practical exercise.

Week	Days	Activities
Week 5	Days 25-28	Project closure: supervisor review and feedback, internship report drafting, presentation preparation and final review.

5. Week-wise Narrative

Week 1: The first week was devoted to orientation and foundations. After company induction, the student was introduced to Artificial Intelligence and studied its types, applications and the difference between AI, ML and DL. The week ended with Python for AI - Python basics, NumPy and Pandas - supported by small coding exercises in a notebook.

Week 2: In the second week the study moved to data preprocessing - cleaning data, handling missing values, encoding and scaling - followed by supervised learning, covering regression and classification with labelled data, and unsupervised learning, covering clustering, dimensionality reduction and pattern finding.

Week 3: The third week focused on model evaluation, including accuracy, precision, recall, the confusion matrix and overfitting; neural network basics, including neurons, layers, activation functions and training; and NLP basics, including text processing, tokenisation and simple language tasks.

Week 4: In the fourth week the student studied computer vision basics - image data, basic image processing and classification - and completed the mini AI project, planning, building and testing a small AI model end to end. A case study exercise and a mock practical exercise were also completed in this week.

Week 5: The last week was devoted to project closure: supervisor review and feedback, drafting of the internship report, preparation of a presentation and final review before submission.

6. Topic-wise Learning in Detail

6.1 AI Fundamentals

Concept: What AI is, its types, applications and the difference between AI, ML and DL. **Practice carried out:** Preparing a comparison chart of AI, ML and DL with examples, listing ten everyday AI applications with the kind of data each uses, and writing a one-page note on one AI use in a local setting.

6.2 Python for AI

Concept: Python basics, NumPy and Pandas used in AI work. **Practice carried out:** Writing programs with loops and functions, creating and reshaping NumPy arrays, calculating statistics, loading a CSV file into Pandas, filtering, grouping and plotting the data.

6.3 Data Preprocessing

Concept: Cleaning data, handling missing values, encoding and scaling. **Practice carried out:** Cleaning a messy student-records table: counting and filling missing values, removing duplicates, one-hot encoding a category column, scaling numeric columns and splitting into train and test sets.

6.4 Supervised Learning

Concept: Regression and classification with labelled data. **Practice carried out:** Fitting a linear regression of marks on study hours, training logistic regression and decision tree models to predict pass or fail, and comparing their test results.

6.5 Unsupervised Learning

Concept: Clustering, dimensionality reduction and pattern finding. **Practice carried out:** Clustering sample customers with k-means, choosing k with the elbow method, reducing a dataset to two dimensions with PCA and describing each cluster in words.

6.6 Model Evaluation

Concept: Accuracy, precision, recall, confusion matrix and overfitting. **Practice carried out:** Building a confusion matrix, calculating the metrics by hand and in code, comparing training and test accuracy to detect overfitting, and using cross-validation.

6.7 Neural Network Basics

Concept: Neurons, layers, activation functions and training. **Practice carried out:** Calculating the output of a single neuron by hand, building a small network with two hidden layers in Keras on a sample dataset, and plotting loss against epochs.

6.8 NLP Basics

Concept: Text processing, tokenisation and simple language tasks. **Practice carried out:** Cleaning and tokenising sentences, removing stop words, building bag-of-words and TF-IDF tables, and training a small sentiment classifier on dummy reviews.

6.9 Computer Vision Basics

Concept: Image data, basic image processing and classification. **Practice carried out:** Loading images as arrays, converting to grayscale, resizing and normalising, applying blur and edge filters, and classifying small images of handwritten digits.

6.10 Mini AI Project

Concept: Planning, building and testing a small AI model end to end. **Practice carried out:** Completing the project described in Section 10 and writing its model card.

7. Rules, Ethics and Good Practice - Points Studied

AI operates within a legal and ethical framework. Personal data, such as a name, phone number, photograph or health record, may be collected and used only with the person's consent and for a stated purpose under the Digital Personal Data Protection Act, 2023. Datasets, code and images must respect the Copyright Act, 1957 and the licence of the source. The Information Technology Act, 2000 governs online conduct and electronic records. Where a model affects people, it should be tested for unfair differences between groups and its decisions should be explainable and open to human review.

Good practice includes using open or dummy data for learning, removing personal identifiers, keeping data and accounts secure, using two-step verification for cloud notebooks and code repositories, recording the data source, version and settings of every experiment, checking AI-generated text and code before use, being open about the limits of a model, and never presenting a prediction as a certainty.

8. Process Flows Studied

Process mapping was used to understand how an AI solution moves from idea to deployment and who is responsible at each step. The main flows studied were:

- **AI project:** Problem and success measure > Data collection > Exploration > Cleaning and preparation > Model building > Evaluation > Improvement > Explanation and report > Monitoring.
- **Data preprocessing:** Load data > Inspect types and shape > Handle missing values > Remove duplicates > Treat outliers > Encode categories > Scale numbers > Split into train and test.
- **Supervised learning:** Define target > Select features > Split data > Fit model > Predict on test data > Evaluate > Tune settings > Final check.
- **Clustering:** Select features > Scale > Choose k > Fit k-means > Check elbow or silhouette > Describe each cluster > Use or revise.
- **Neural network training:** Prepare data > Define layers > Choose loss and optimiser > Forward pass > Calculate loss > Backpropagation > Update weights > Repeat for epochs > Evaluate.

- **NLP and vision pipelines:** Collect text or images > Clean or resize > Convert to numbers > Train > Evaluate > Review errors.

9. Worked Examples with Sample (Dummy) Data

The examples below use a fictitious local business, Gyan Jyoti Classes, a coaching centre for Class 9 and 10 in a town in Bihar. All names, numbers and results are dummy figures for learning.

Example A: Framing the problem

The centre wants to identify, early, the Class 10 students who are at risk of failing the pre-board test, so that teachers can give extra help. The task is therefore *binary classification*: the label is 'at risk' or 'not at risk'. The dummy dataset has 500 students with the features attendance percentage, average weekly test score, daily study hours, class and medium of instruction. Success is measured mainly by recall (finding as many at-risk students as possible) with a reasonable precision, because a missed student is a bigger loss than an unnecessary extra class. Predictions are used only as a guide, and a teacher takes the final decision.

Example B: Data preprocessing

The raw file had 512 rows; 12 duplicate rows were removed, leaving 500. The attendance column had 18 missing values (18 / 500 = 3.6%), which were filled with the median. In a small check column the marks were 45, 60, (missing), 72 and 58. The mean of the four known values is (45 + 60 + 72 + 58) / 4 = 235 / 4 = 58.75, so the missing mark could be filled with 58.75. Min-max scaling of the mark 60, with minimum 45 and maximum 72, gives (60 − 45) / (72 − 45) = 15 / 27 = about 0.556. The column 'medium' (Hindi or English) was one-hot encoded into two 0/1 columns. The 500 rows were split 80:20 into 400 for training and 100 for testing, and the filling values and scaler were learned from the 400 training rows only, to avoid leakage.

Example C: Python, NumPy and Pandas

```
import numpy as np
import pandas as pd

scores = np.array([45, 60, 72, 58, 81])
print(scores.mean())      # 63.2
print(np.median(scores))  # 60.0
print(scores.std())       # about 12.35 (population standard deviation)

df = pd.read_csv("students_dummy.csv")
print(df.shape)           # (500, 6)
print(df.isnull().sum())  # missing values per column
print(df.groupby("medium")["test_avg"].mean())
```

The mean is (45 + 60 + 72 + 58 + 81) / 5 = 316 / 5 = 63.2. The median is the middle sorted value, 60. The squared deviations from the mean add up to 762.8, so the variance is 762.8 / 5 = 152.56 and the standard deviation is its square root, about 12.35.

Example D: Regression and classification

A linear regression of marks on daily study hours used the five points (1, 35), (2, 45), (3, 50), (4, 62) and (5, 68). The mean of x is 3 and the mean of y is 52. The sum of (x − 3)(y − 52) is 83 and the sum of (x − 3)² is 10, so the slope m = 83 / 10 = 8.3 and the intercept c = 52 − 8.3 × 3 = 27.1. The line is marks = 27.1 + 8.3 × hours, which predicts 76.9 marks for 6 hours (6 is just outside the data range, so the prediction should be treated with caution). The sum of squared errors is 9.10, giving MSE = 1.82, RMSE about 1.35 and $R^2 = 1 − 9.10 / 698$ = about 0.987 on this tiny sample.

For the at-risk classification the simplest baseline, predicting 'not at risk' for everyone, scored 55% on the 100 test students (55 of them were not at risk). Logistic regression scored 85% and a decision tree scored 81%, so logistic regression was chosen as the first model.

Example E: Clustering

Six customers of a stationery shop spend Rs. 200, 250, 300, 1,200, 1,300 and 1,400 a month. Starting k-means with k = 2 and the centres 200 and 1,400, the points 200, 250 and 300 join the first centre and 1,200, 1,300 and 1,400 join the second. The new centres are 250 and 1,300, and a second pass changes nothing, so the algorithm stops. The total within-cluster squared distance is 5,000 + 20,000 = 25,000, compared with 1,678,750 if all six are kept in a single cluster, a very large fall that supports k = 2 in the elbow method. The clusters were named 'low spenders' and 'high spenders' and described in words.

Example F: Evaluating the classifier

On the 100 test students the confusion matrix was TP = 40, FP = 10, FN = 5 and TN = 45. Accuracy = (40 + 45) / 100 = 85%. Precision = 40 / (40 + 10) = 80%. Recall = 40 / (40 + 5) = about 88.9%. F1 = 2 × 0.80 × 0.889 / (0.80 + 0.889) = about 84.2%. Of 45 students who really were at risk, 40 were found, and of 50 students flagged, 40 really were at risk. A decision tree with no depth limit scored 98% on the training data but only 78% on the test data, a gap of 20 points that shows overfitting; limiting its depth reduced the gap.

Example G: A single neuron and a small network

A neuron has inputs x_1 = 2 and x_2 = 3, weights 0.5 and −0.2 and bias 0.1. Then z = (0.5 × 2) + (−0.2 × 3) + 0.1 = 1.0 − 0.6 + 0.1 = 0.5. With a sigmoid activation the output is 1 / (1 + e^−0.5) = about 0.622; with ReLU the output is 0.5. A small network with 3 inputs, a hidden layer of 8 neurons, a second hidden layer of 4 and one output has (3×8 + 8) + (8×4 + 4) + (4×1 + 1) = 32 + 36 + 5 = 73 trainable parameters.

Example H: Text processing

The sentence 'AI is changing how India learns' becomes six tokens after lower-casing and splitting: ai, is, changing, how, india, learns. Removing the stop words 'is' and 'how' leaves four tokens: ai, changing, india, learns. For a bag-of-words table with the vocabulary [good, course, teacher, bad], the review 'good course good teacher' is the vector [2, 1, 1, 0] and the review 'bad course' is [0, 1, 0, 1]. A small sentiment classifier trained on dummy reviews classified 44 of 50 test reviews correctly, an accuracy of 88%.

Example I: Image data

A 28 × 28 grayscale image has 784 pixels, each from 0 to 255; a pixel of 128 becomes 128 / 255 = about 0.502 after normalisation. A colour image of 224 × 224 pixels has 224 × 224 × 3 = 150,528 values. For the 3 × 3 patch with rows (10, 20, 30), (40, 50, 60) and (70, 80, 90), an averaging filter gives (10 + 20 + … + 90) / 9 = 450 / 9 = 50 for the centre pixel. A small classifier of handwritten-digit images scored 920 correct out of 1,000 test images, or 92%.

10. Project: AI Mini Project using Machine Learning

Scope. The internship work was organised around a project in which data was prepared, a small machine learning model was built and tested, and a performance report was written, using an open or dummy dataset and free tools.

Steps followed. (i) Defined the problem, the target and the success measure; (ii) loaded and explored the dataset with Pandas and charts; (iii) cleaned the data, handled missing values, encoded categories and scaled numbers; (iv) split the data into training and test sets; (v) built a baseline, a logistic regression and a decision tree with scikit-learn; (vi) evaluated them with a confusion matrix, precision, recall and F1, checked for overfitting and compared the models; (vii) wrote the findings, limits and a model card, and reviewed the work with the supervisor.

Skills and tools. Python, NumPy, Pandas, Matplotlib, scikit-learn, Jupyter Notebook or Google Colab, and MS Excel for summary tables.

Outputs. Problem statement, cleaned dataset description, exploration notebook, model comparison table, confusion matrix and metric sheet, model card, presentation, and this internship report.

11. Case Study Exercise: A Model That Was 95% Accurate and Still Failed

A case study exercise was completed on a typical AI problem. The business and figures are illustrative and not taken from any real organisation.

Background. A small finance company (fictitious) built a model to predict which of its borrowers would fail to repay a loan. The data had 1,000 past borrowers, of whom 950 repaid and 50 defaulted. The trainee reported with pride that the model was 95% accurate.

Problem observed. On checking the confusion matrix it was found that the model predicted 'will repay' for every borrower. It was therefore right for the 950 who repaid and wrong for all 50 who defaulted, so accuracy was 950 / 1,000 = 95% but recall for defaulters was 0 / 50 = 0%. The model had learned nothing useful. The data had also been split without keeping the same share of defaulters in each part, no baseline had been compared, and the features included a column that was filled in only after a loan went bad, which leaked the answer into the training data.

Analysis. The main problem was not the algorithm but the choice of metric and the handling of the data. With a rare class, accuracy hides failure. The leaking column made early results look better than reality would be, and the missing baseline hid the fact that 'always predict repay' already scored 95%.

Recommendations. (1) Judge the model by recall, precision, F1 and the precision-recall curve for the defaulter class, not by accuracy alone. (2) Use a stratified train-test split. (3) Remove features that are not available at the time of the decision. (4) Try class weights, resampling and a lower decision threshold to find more defaulters. (5) Compare every model with a simple baseline. (6) Test whether error rates differ across groups of borrowers, and keep a human officer in the loop for decisions that affect people. (7) Monitor the model after use, because customer behaviour changes.

Expected result and lesson learned. With the right metric, clean features and balanced handling of the rare class, the model is expected to find a meaningful share of defaulters at an acceptable false-alarm rate. The lesson is that AI is a system: data, features, metrics, testing and human judgement must work together, and numbers must be questioned before they are trusted.

12. Mock Practical Exercise: An End-to-End Mini AI Pipeline

In the mock practical exercise the student planned and completed a small AI pipeline on a sample (dummy) dataset, from the first question to the final report, using the tools and methods practised earlier. The steps followed were:

1. Problem: wrote the question, the target, the success measure and the risks.
2. Data: loaded the dataset with Pandas, noted its source and checked rows, columns and types.
3. Exploration: drew histograms and a correlation table and listed data problems.
4. Preparation: handled missing values and duplicates, encoded categories, scaled numbers and split into train and test sets.
5. Models: built a baseline, a regression or classification model, and a clustering of the same data.
6. Evaluation: calculated accuracy, precision, recall, F1 and a train-versus-test comparison, and examined wrong predictions.
7. Extensions: tokenised a few text lines and processed a small image to show how text and images become numbers.
8. Reporting: wrote a one-page summary with results, limits, ethical points and next steps.

The exercise showed that every step depends on the one before it: wrong data leads to a wrong model, a leaked feature gives false confidence, a missing baseline hides weak results, and an honest report is the base of the next improvement.

13. Sample Checklists Prepared

Checklists were prepared as part of the hands-on tasks. They reduce errors and make routine work easier to review.

Data Preparation Checklist

- Source, licence and permission for the data are noted; personal identifiers removed.
- Shape, column types and summary statistics checked.
- Missing values, duplicates and outliers examined and handled with a reason recorded.
- Categories encoded and numbers scaled where needed.
- Train and test sets split before any learning step; stratified if a class is rare.
- Raw data kept unchanged; cleaned data saved separately.

Model Building Checklist

- Target and features clearly defined; no feature leaks the answer.
- A simple baseline built first.
- Random seed fixed so results can be repeated.
- Settings (hyper-parameters) recorded for each run.
- Cross-validation used where data is small.

Model Evaluation Checklist

- Metric chosen to match the cost of errors.
- Confusion matrix, precision, recall and F1 calculated for classification; MAE or RMSE for regression.
- Training and test scores compared for overfitting.
- Wrong predictions read and grouped.
- Performance compared across groups of people, if relevant.

Neural Network Checklist

- Inputs scaled; layers, activations and loss function suit the task.
- Learning rate, epochs and batch size recorded.
- Training and validation loss plotted; early stopping or dropout used if validation loss rises.

NLP and Vision Checklist

- Text cleaned, tokenised and converted to numbers in the same way for training and testing.
- Images resized to the same size and pixel values normalised.
- Consent and privacy considered before using any personal text or photograph.

Report and Ethics Checklist

- Problem, data, method, results and limits stated clearly and honestly.
- Sources and libraries credited.
- Model card written; intended use and misuse stated.
- Notebook runs from top to bottom without errors.

14. Tools and Resources

Learning materials, Python, Jupyter Notebook and Google Colab, the libraries NumPy, Pandas, Matplotlib and scikit-learn, Keras/TensorFlow for neural network basics, NLTK for text processing, OpenCV and Pillow for images, open datasets from sources such as Kaggle and the UCI Machine Learning Repository (used according to their licences), MS Excel for summaries, MS Word and PowerPoint for documentation and presentation, official library documentation, assignments and mentor feedback were used during the

internship.

15. Programme at a Glance

The programme was in the domain of Artificial Intelligence with RKNEXORA PRIVATE LIMITED as the agency. It required a minimum of 120 hours over 28 learning days, organised in three phases: Orientation (Days 1-2), Learning and Practice (Days 3-22) and Project and Review (Days 23-28). Attendance was recorded in the Internship Attendance Log, signed by the supervisor (see Attendance Sheet).

LEARNING OUTCOME

a. Knowledge Acquired

1. What AI is, its types and applications, and how AI, machine learning and deep learning differ.
2. How to use Python, NumPy and Pandas to load, explore and summarise data.
3. How to clean data, handle missing values, encode categories, scale numbers and split data correctly.
4. How supervised learning (regression and classification) and unsupervised learning (clustering and PCA) work.
5. How to evaluate models with a confusion matrix, accuracy, precision, recall and F1, and how to recognise overfitting.
6. How neurons, layers, activation functions and training work in a neural network.
7. How text and images are converted into numbers for NLP and computer vision, and how to write and present a short AI project report responsibly.

Glossary of Terms Learned

- **AI** - Artificial Intelligence: machines performing tasks that need human-like intelligence.
- **ML / DL** - Machine Learning / Deep Learning: learning from data / learning with many-layered neural networks.
- **Dataset / Feature / Label** - A collection of data / an input column / the answer to be predicted.
- **Training set / Test set** - Data used to learn / data kept aside to check the model.
- **Supervised / Unsupervised learning** - Learning with labels / finding structure without labels.
- **Regression / Classification** - Predicting a number / predicting a category.
- **Clustering / PCA** - Grouping similar items / reducing the number of features.
- **Overfitting / Underfitting** - Model too closely fitted to training data / too simple to learn the pattern.
- **Data leakage** - Information from the answer or test data wrongly used in training.
- **Confusion matrix** - Table of TP, FP, FN and TN counts.
- **Precision / Recall / F1** - Correctness of positive predictions / share of real positives found / their balance.
- **Hyper-parameter** - A setting chosen before training, such as k or tree depth.
- **Neuron / Activation function** - A unit that sums weighted inputs / a function such as ReLU or sigmoid applied to the sum.
- **Epoch / Learning rate** - One pass over the data / the size of each weight update.
- **Tokenisation / TF-IDF** - Splitting text into units / a word-importance score.
- **Pixel / Filter** - One value of an image / a small grid used to process images.
- **Bias (in data)** - Unfair patterns in data that lead to unfair results.
- **Model card** - A short document describing what a model does, its data, performance and limits.

b. Skills Learned

Technical skills. Writing Python programs; using NumPy arrays and Pandas tables; plotting charts; cleaning, encoding and scaling data; training regression, classification and clustering models with scikit-learn; calculating and interpreting evaluation metrics; building a small neural network; tokenising text and building bag-of-words tables; processing images as arrays; and writing a project notebook and model card.

Professional skills. Logical and analytical thinking; problem solving; clear writing and explanation of technical ideas in simple words; attention to detail and debugging; time management; ethical and legal

awareness; teamwork and acceptance of feedback; and documentation, report writing and presentation.

These skills were developed through repeated practice and daily documentation. Coding and metric work improved analytical thinking, debugging improved patience and attention to detail, comparing models improved a habit of testing before deciding, and writing remarks and reports improved communication.

c. Observed Attitude and Gained Values

1. Problem first: AI is useful only when it solves a real and clearly stated problem.
2. Honesty and transparency: results, limits and mistakes must be reported truthfully.
3. Respect for privacy and consent: data about people belongs to those people.
4. Fairness and responsibility: models must be checked for harm, and people stay accountable for decisions.
5. Curiosity and discipline: tools change quickly, so regular learning, practice and punctual documentation are essential.

d. Challenges Faced

Understanding terms and ideas such as overfitting, gradient descent, precision and recall, and TF-IDF was handled by using a glossary, small hand calculations and mentor guidance.

Learning several libraries in a short time was handled by practising each one on a small task the same day and keeping short notes of useful commands.

Debugging errors and unexpected results, such as shape mismatches or very high accuracy caused by leakage, was handled by reading error messages carefully, printing intermediate results and checking with the supervisor.

Working without large datasets and powerful computers was handled by using small open datasets, free cloud notebooks and clear assumptions written beside each result.

Maintaining consistent documentation was handled by using daily log entries and reviewing feedback to improve documentation progressively.

Managing time between practice, project work and report writing was handled by dividing work week by week and using review days for corrections.

e. Career Pathways

The knowledge gained supports entry-level roles and further learning towards careers such as AI or Machine Learning Engineer, Data Analyst, Data Scientist, NLP Engineer, Computer Vision Engineer, Business Intelligence Analyst, AI Research Assistant, MLOps Engineer and AI Product or Project Associate. Opportunities exist in IT services and start-ups, banking and finance, healthcare, education, agriculture, e-commerce, government and research organisations, and as a freelancer or consultant. Related study options include BCA, B.Tech, MCA, M.Sc. or M.Tech in Computer Science, Data Science or Artificial Intelligence, and online courses such as those offered through NPTEL/SWAYAM, Coursera and Kaggle Learn, as well as entry-level AI and cloud certifications. Because tools and models change quickly, continuous learning and a portfolio of real projects are essential.

CONCLUSION

The internship in Artificial Intelligence was a valuable learning experience that integrated academic knowledge with practical activity. The programme achieved its objectives through structured learning, guided practice, assignments, supervisor review and documentation.

The student gained a clear understanding of how a small AI solution is planned, built, tested and explained - through AI fundamentals, Python for AI, data preprocessing, supervised and unsupervised learning, model evaluation, neural networks, NLP and computer vision - and acquired practical familiarity with notebooks, datasets, models, metrics and reports. The experience has strengthened the student's employability and provides a foundation for further study and work in computing, data and analytics.

Reflection

Looking back, the most useful lesson of the internship was that AI is a chain, and a chain is only as strong as its weakest link. A clear problem points to the right data, clean data makes learning possible, a suitable model finds the pattern, honest evaluation shows whether it really works, and a careful report shows what to do next. The case study showed how a model that looked 95% accurate could be completely useless. Moving from concept to practice to documentation made abstract ideas concrete, and supervisor feedback showed where understanding was weak. The habit of recording each day's work also made it easier to see progress over the weeks.

Objectives and Achievement

The objective of understanding AI fundamentals was achieved through concept sessions and a comparison chart of AI, ML and DL. The objective of learning Python, NumPy and Pandas was achieved through coding exercises and a data exploration notebook. The objective of learning data preprocessing was achieved through cleaning a messy dataset and preparing train and test sets. The objective of learning supervised and unsupervised learning was achieved through regression, classification and clustering exercises. The objective of learning model evaluation, neural networks, NLP and computer vision was achieved through metric calculations, a small network, text processing and image processing tasks. The objective of completing a mini AI project was achieved through the end-to-end project and model card. The objective of developing professional discipline was achieved through daily attendance, supervisor review and report writing.

Future Scope

The knowledge gained can support further study of computer science, data science and artificial intelligence, and careers in IT, finance, healthcare, education, agriculture and public services. With growing use of AI in everyday tools, regional-language technology, automation and generative AI, trained and ethical AI practitioners are increasingly needed, including in Bihar.

Suggestions for Improvement of the Programme

1. Include live coding practice on larger open datasets and talks by AI professionals and industry experts, with appropriate permissions.
2. Continue regular mentor feedback and progress reviews.
3. Provide more project-based activities such as solving a small real problem for a local organisation under supervision.
4. Encourage students to maintain detailed daily documentation and to share their notebooks for review.

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ATTENDANCE SHEET

The Internship Attendance Log issued by the IPO, with the supervisor's signature for each day, is to be attached on this page.

[Attach the signed Internship Attendance Log here]